

EX 7

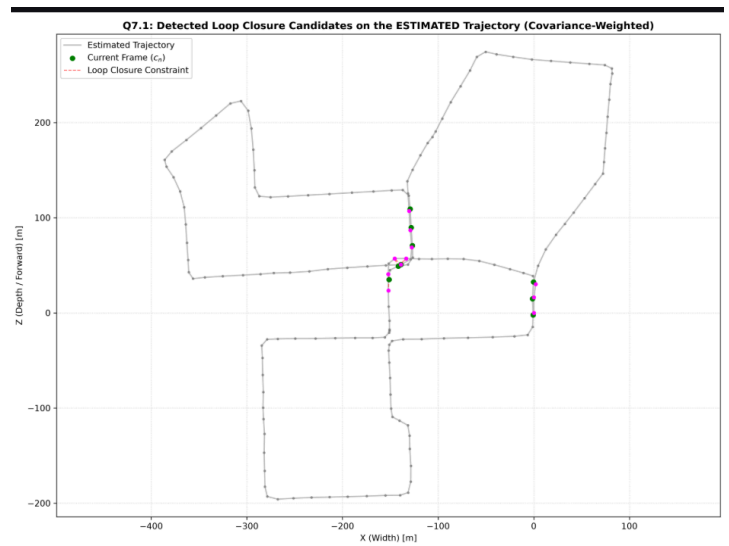
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[Github](#)

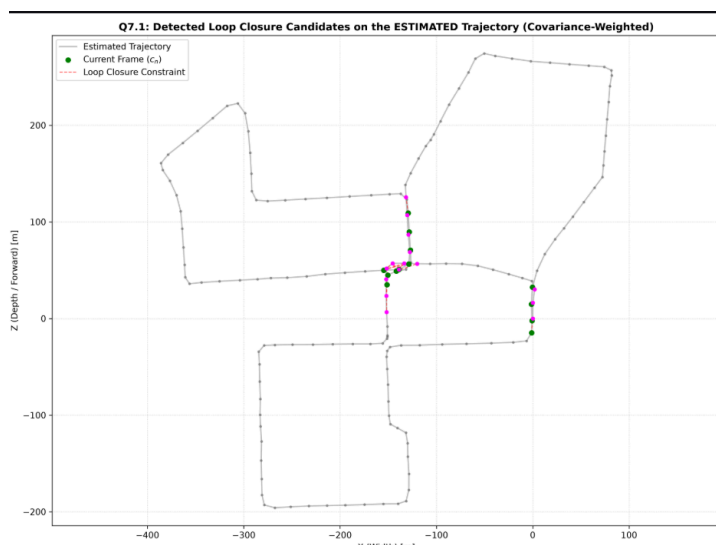
threshold 100:



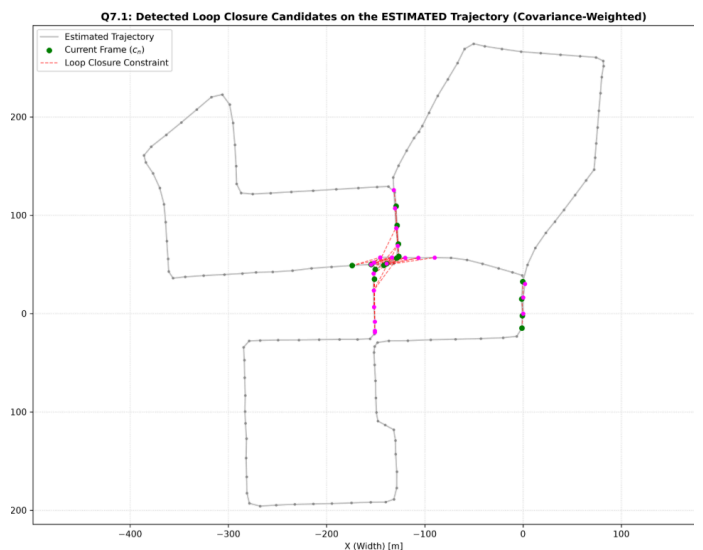
threshold 1000:



threshold 2000:



threshold 5000:



A)

To estimate the relative covariance $\Sigma_{n|i}$ between a current pose c_n and a past candidate pose c_i , a covariance-weighted topological pose graph was constructed using the networkx library.

- **Edge Weights Selection:** The edges connecting sequential odometry nodes were weighted using the trace of each relative covariance matrix (np.trace), which sums the variances across all 6 degrees of freedom (3 translation + 3 rotation) to provide an overall scalar measure of positional and rotational uncertainty. Under this framework, the chosen path is considered the "shortest" path in a statistical sense — minimizing the total accumulated structural noise rather than physical metric distance. Alternative reasonable choices for edge weights could include the determinant of the covariance matrix (representing the physical volume of the uncertainty ellipsoid) or the maximum eigenvalue (isolating worst-case directional variance).
- **Shortest Path Implementation:** The shortest path between any keyframe pair was efficiently resolved using Dijkstra's algorithm via `networkx.shortest_path` based on the assigned trace weights. Once the optimal path was resolved, the compound relative covariance $\Sigma_{\square|i}$ was compiled by walking the path edge by edge and, for each edge, transforming its covariance (originally expressed in that edge's own local frame) into a shared reference frame via the SE(3) Adjoint map of the remaining transform to c_i , before summing the transformed covariances. This frame-alignment step is necessary because each edge's covariance is only valid in its own local coordinate frame; naively summing the raw covariance matrices without first rotating them into a common frame would not correctly propagate uncertainty through a chain of composed relative poses (the standard first-order "compounding" rule for chained SE(3) transforms).

B)

Once the shortest path and relative covariance $\Sigma_{\square|i}$ were established, potential loop closure candidates were isolated by applying a rigorous Mahalanobis distance test:

$$\text{DistM} = \Delta c_{\square|i}^T \Sigma_{\square|i}^{-1} \Delta c_{\square|i}$$

where $\Delta c_{\square|i}$ represents the accumulated relative pose parameterized into a 6-dimensional error vector via the SE(3) Logmap operation. To balance true-positive loop recall against downstream execution complexity, the threshold was evaluated at two distinct levels, **100 (Left Image)** and **1000 (Right Image)**:

- **Threshold = 100 (Left Image):** Represents an overly restrictive baseline. While it isolates a highly precise subset of candidates along the central vertical axis, it suffers from severe under-mapping due to the heavy odometry drift accumulated over the long sequence. By neglecting tracking uncertainty, this conservative threshold completely misses critical loop closure opportunities, such as the vital return loop near the origin.
- **Threshold = 1000 (Right Image — Chosen Configuration):** Establishes the optimal equilibrium for the framework. By introducing the correct amount of statistical leniency, it successfully compensates for the inflated uncertainty bounds ($\Sigma_{\square|i}$) caused by long-term drift. This configuration successfully uncovers all valid geometric

intersections — most notably reclaiming the essential true-positive loops within the high-drift zone near the origin (bottom-right). This ensures a comprehensive topological map while keeping the candidate pool clean enough to be efficiently validated by the subsequent feature-matching stage.

- **Threshold = 2000 & Threshold = 5000:** Increasing the threshold beyond the optimal configuration of 1000 introduces excessive statistical leniency, progressively leading to the inclusion of noisy and incorrect constraints (False Positives). The onset of this trend can already be observed to a minor extent in the Threshold = 2000 plot, which primarily exhibits high density and local noise among constraints along the central vertical axis. Conversely, in the Threshold = 5000 plot, this phenomenon becomes highly pronounced. Due to the overly inflated uncertainty bounds, the framework begins to generate completely invalid loop closure constraints. These constraints stretch horizontally, erroneously linking geometrically distant trajectory segments and pathways that do not share an actual spatial overlap—ultimately risking severe distortion of the trajectory optimization rather than correcting it.

Ultimately, a threshold of 100 suffers from premature filtering and poor recall, whereas the subsequent expansion to 2000 and 5000 introduces excessive noise and invalid spatial cross-connections. Consequently, the chosen threshold of 1000 establishes the ideal equilibrium—maximizing mapping completeness and providing a robust, drift-compensated foundation for visual verification without compromising geometric integrity

Hence:

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RUNNING: Q7.2 - CONSENSUS MATCHING (VISUAL VERIFICATION)
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[Q7.2] Testing Link c_1440 -> c_0: Total Matches = 537 | Inliers = 464 | Ratio = 86.41% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 86.41%, Inliers: 464
[Q7.2] Testing Link c_1460 -> c_20: Total Matches = 518 | Inliers = 447 | Ratio = 86.29% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 86.29%, Inliers: 447
[Q7.2] Testing Link c_1480 -> c_40: Total Matches = 1588 | Inliers = 1468 | Ratio = 92.39% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 92.39%, Inliers: 1468
[Q7.2] Testing Link c_2300 -> c_240: Total Matches = 320 | Inliers = 287 | Ratio = 89.69% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 89.69%, Inliers: 287
[Q7.2] Testing Link c_2300 -> c_260: Total Matches = 243 | Inliers = 218 | Ratio = 89.71% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 89.71%, Inliers: 218
[Q7.2] Testing Link c_3160 -> c_2220: Total Matches = 405 | Inliers = 352 | Ratio = 86.91% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 86.91%, Inliers: 352
[Q7.2] Testing Link c_3180 -> c_2240: Total Matches = 229 | Inliers = 202 | Ratio = 88.21% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 88.21%, Inliers: 202
[Q7.2] Testing Link c_3200 -> c_2260: Total Matches = 293 | Inliers = 232 | Ratio = 79.18% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 79.18%, Inliers: 232
[Q7.2] Testing Link c_3260 -> c_240: Total Matches = 176 | Inliers = 157 | Ratio = 89.20% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 89.20%, Inliers: 157
[Q7.2] Testing Link c_3260 -> c_260: Total Matches = 445 | Inliers = 411 | Ratio = 92.36% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 92.36%, Inliers: 411
[Q7.2] Testing Link c_3260 -> c_2300: Total Matches = 389 | Inliers = 366 | Ratio = 94.09% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 94.09%, Inliers: 366
[Q7.2] Testing Link c_3299 -> c_300: Total Matches = 694 | Inliers = 607 | Ratio = 87.46% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 87.46%, Inliers: 607
[Q7.2] Testing Link c_3299 -> c_320: Total Matches = 44 | Inliers = 34 | Ratio = 77.27% (Required Ratio: 75.0%, Min Inliers: 20)
--> [SUCCESS] Loop Verified! Ratio: 77.27%, Inliers: 34

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SUMMARY Q7.2: Total Visually Verified Loops: 13
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7.2:

To geometrically verify the loop closure candidates, we implemented a **dual-restraint filtering criterion** that enforces two threshold conditions simultaneously:

1. Inlier Ratio Threshold of 0.75 (75%):

- Building upon the lecture guidelines which establish a 60% baseline for high-quality local matching, we elevated this threshold to a more conservative 75% to optimize the global pose graph optimization stage. Because loop closures introduce global trajectory constraints where a single false positive can significantly distort the entire map, this higher threshold reinforces structural safety.

2. Minimum Absolute Inlier Count of 20 Points:

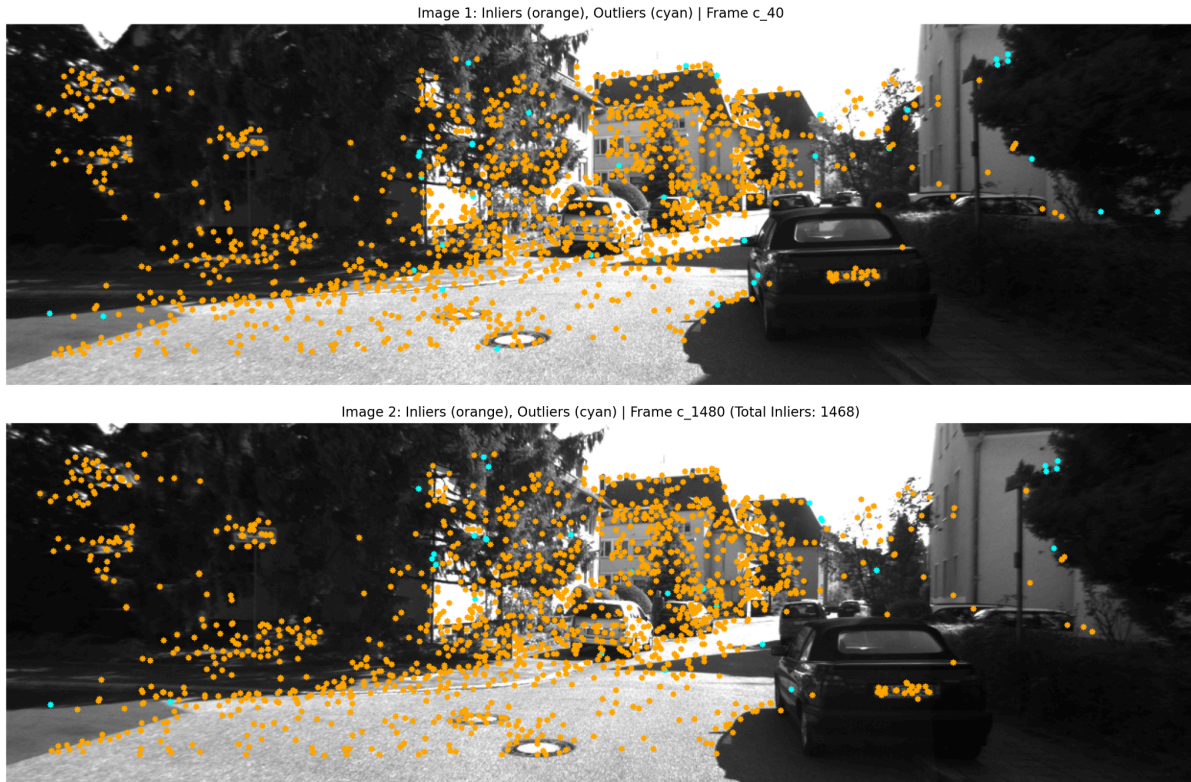
- The inlier ratio is a relative metric and can be misleadingly high in feature-sparse scenes (e.g., getting 4 out of 5 matches yields an 80% ratio). However, a low absolute number of points lacks the necessary degrees of freedom for a stable downstream Bundle Adjustment. Setting a hard lower bound of 20 inliers guarantees sufficient geometric constraints to extract a reliable relative pose and covariance.

Conclusion: Only frame pairs that passed both constraints concurrently were accepted as verified loop closures. This dual-restraint approach effectively eliminated catastrophic optimization errors while enabling accurate global drift correction.



7.3:

A local bundle adjustment was performed using the verified inlier correspondences between two keyframes. The first camera pose was fixed with a prior factor, the second pose was initialized using the PnP-RANSAC estimate, and the 3D landmarks were initialized by stereo triangulation from the first keyframe. After optimization, the relative covariance was extracted from the marginal information matrix by inverting the $6 \times 66 \times 66 \times 6$ information block corresponding to the optimized second pose, and was later used as the noise model for the loop-closure constraint.



7.4:

Each successful loop closure was added to the global pose graph as a `BetweenFactorPose3` using the optimized relative pose and its corresponding covariance. The graph was re-optimized after every added constraint, using the previous optimized solution as the initial estimate. Since the bundle adjustment estimated the transformation in the opposite direction of the graph edge, the inverse relative pose was inserted into the pose graph.

7.5:

A total of **13 successful loop closures** were detected and added to the pose graph. Figure X shows one successful consensus match with inliers and outliers. To illustrate the optimization process, we present four pose graph snapshots: before adding loop closures, after the first loop closure, after approximately half of the loop closures, and after all loop closures were added, together with their location covariance ellipses. The optimized trajectory is

compared to the ground truth both with and without loop closures, showing a significant reduction in accumulated drift. The absolute location error plot demonstrates consistently lower localization error after adding loop closures, while the location uncertainty plot shows a substantial decrease in uncertainty throughout most of the

trajectory. The uncertainty size was measured as $\sqrt{\det(\Sigma_{XZ})}$, where Σ_{XZ} is the 2D marginal covariance of the camera position in the X-Z plane, which is proportional to the area of the corresponding uncertainty ellipse.

